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# Customization Builder

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This is the official reference for the features and functionality of the Customization Experience Builder, a product in the [Customization Experience Platform \(CXP\)](https://nde-devportal-docs.niketech.com/doc/commerce/customization/overview-customization.html) ( <https://nde-devportal-docs.niketech.com/doc/commerce/customization/overview-customization.html> ).

**TIP:** Also see [Customization Overview](https://nde-devportal-docs.niketech.com/doc/commerce/customization/overview-customization.html) ( <https://nde-devportal-docs.niketech.com/doc/commerce/customization/overview-customization.html> ) and [Adding Customization To Your Experience](https://nde-devportal-docs.niketech.com/doc/commerce/customization/use-customization.html) ( <https://nde-devportal-docs.niketech.com/doc/commerce/customization/use-customization.html> ).

The Builder is a JavaScript bundle that is your main interface with CXP. It offers the following:

- **UX:** Returns a fully-styled UX for customizing products
- **Data API:** Interact with [Build Data](#) and CXP REST APIs

## Basic Integration

This section describes how to complete a basic integration of the Builder into a web view or browser-based application.

**TIP:** For more detailed integration instructions see [Adding Customization To Your Experience](https://nde-devportal-docs.niketech.com/doc/commerce/customization/use-customization.html) ( <https://nde-devportal-docs.niketech.com/doc/commerce/customization/use-customization.html> ).

### 1. Include the Builder Bundle

In your app’s source (e.g. in an HTML template), include the Builder JavaScript bundle in a `<script>` tag in the `<body>` like:

```
<script src="https://assets.commerce.nikecloud.com/nikeid/builder/dist/b16Builder.bundle.min.js" type="text/javascript"></script>
```

This loads the Builder bundle when the page is rendered.

### Bundle URIs

The bundle URIs to be used in the Test and Production environments are provided below. Reach out to the CXP team to determine which version you should be using.

#### Test

- Builder UI: [https://assets.test.commerce.nikecloud.com/nikeid/builder/dist/b16Builder.\[bundleVersion\].min.js](https://assets.test.commerce.nikecloud.com/nikeid/builder/dist/b16Builder.[bundleVersion].min.js)
- Product API: [https://assets.test.commerce.nikecloud.com/nikeid/builder/dist/b16ProductApi.\[bundleVersion\].min.js](https://assets.test.commerce.nikecloud.com/nikeid/builder/dist/b16ProductApi.[bundleVersion].min.js)

#### Production

- Builder UI: [https://assets.commerce.nikecloud.com/nikeid/builder/dist/b16Builder.\[bundleVersion\].min.js](https://assets.commerce.nikecloud.com/nikeid/builder/dist/b16Builder.[bundleVersion].min.js)
- Product API: [https://assets.commerce.nikecloud.com/nikeid/builder/dist/b16ProductApi.\[bundleVersion\].min.js](https://assets.commerce.nikecloud.com/nikeid/builder/dist/b16ProductApi.[bundleVersion].min.js)

### 2. Add a `<div>` for the Builder to Load Into

Add a `<div>` in the `<body>` with an `id="nikeid-app"` attribute like:

```
<div id="nikeid-app-container" style="width: 1000px; height: 800px;">
  <div id="nikeid-app">
    </div>
  </div>
```

You can name the `id` attribute however you want, just be sure to use the same name in the next step.

### 3. Initialize the Builder

Initialize the Builder by invoking the `nikeIdBuilder(rootElement, config)` function in a `<script>` tag in the `<body>` like:

```
<script>
const builderApi = nikeIdBuilder(document.getElementById('nikeid-app'), {
  'nike-api-caller-id': 'com.nike:commerce.b16.localhost',
  pathName: 'KobeAD2exoFA18'
})
</script>
```

This returns the [Builder API](#) and renders the Customization UX into the specified HTML element.

TIPS:

- Make sure to use the same id used in Step 2 in the document.getElementById() argument.
- The nikeIdBuilder function is described in detail in [Builder API](#)

## 4. Navigate to the URI to Launch the Experience

- Serve your app/page locally by running npm start or yarn start.
- Launch the Customization experience by navigating to the URI with at least the ?pathName query parameter. The value in ?pathName must match what you set in the pathName property when the Builder was initialized, like:

http://localhost:3000/?pathName=KobeAD2exoFA18

## Additional Query Params

The following optional query parameters can be used in the URI to affect how the Customization experience is rendered.

- imageSize: sets the size of images returned by the Builder in pixels (max 666)
- imageQuality: sets the quality (as bit-depth) of the PNG images returned by the Builder (8/24)
- hideMenu: shows/hides the harnessMenu on page load (true/false)

Example:

http://localhost:3000/?imageSize=666&imageQuality=8&hideMenu=true&pathName=metcon20161601

## Integration Flow

- Invoke nikeIdBuilder(rootElement, config) to get the Builder API. The returned object also contains a method for api.onApiReady, which indicates that the Builder has loaded.
- Listen to price change, analytic, and “done” events coming from the Builder.
- For buying tools, use setSizeType and setSizeAnswer for answering the size-related questions. Use setAnswer for answering Gender and Width questions.
- Invoke setBuild for loading a new build by prebuild id, metric id, or raw build data. This is mainly meant to “reset” the Builder.

# Builder API

Invoking the nikeIdBuilder(rootElement, config) function returns the Builder API.

Usage:

```
/**
 * Function nikeIdBuilder returns the Builder API
 * @param {HTMLElement} rootElement The DOM element in which to inject the Builder
 * @param {Object} config The configuration properties that identify which Builder instance to be returned
 * @return {Object} Returns the exposed builderApi handlers
 */
var rootElement = document.getElementById('my-app-location');
var config = {
  'nike-api-caller-id': 'XXXXXX',
  locale: 'en_US',
  country: 'US',
  pathName: 'afIHigh14_July',
  bridge: {
    // ...
  }
  // ... etc, documented below
};

var builderApi = nikeIdBuilder(rootElement, config)
```

## Parameters

### rootElement

The argument for the rootElement parameter must be an HTML element target to mount the Builder application into.

Example: document.getElementById('nike-id-builder-app');

config

The argument for the `config` parameter must include the configuration properties that identify which Builder instance to be returned.

At minimum, the `pathName` and `nike-api-caller-id` properties are required in order to initialize the Builder.

```
{
  // required properties
  'nike-api-caller-id': 'com.nike:test',
  pathName: 'af1High14_July',

  // sometimes required properties
  productId: 'PROD359583',
  country: 'US',
  locale: 'en_US',

  // suggested bridge
  bridge: {
    onAnalyticsEvent: function() {},
    onApiReady: function(api) {}, // api ready not build
    onDone: function(buildData) {},
    onError: function(error){},
    onPriceUpdate: function(priceData) {},
    onProductLoad: function(buildData) {} // build ready
  }

  // optional properties
  abTestCookieName: null,
  builderMode: 'default',
  currentEnv: 'PROD',
  dataAsStrings: false,
  disablePalette: false,
  eMerchId: 123456,
  enableAnimationInit: false,
  fetchCapacity: false,
  fitToContainer: document.getElementById('nikeid-app-container'),
  hasTouch: true,
  isColorMatching: false,
  metricId: '42',
  prebuildId: '42',
  sizeTypeRegion: 'US',
  myDesignsEnabled: true,
  myDesignCapacity: 20,

  // optional skip price info
  skipPriceInfo: false,
  colorCode: '994',
  styleCode:'AA1617',

  // optional image quality
  imageQuality: 24,
  imageSize: 300,
  viewPortSize: 'S',
}
```

Required Properties

Table 1: Required Build Properties

|          |        |                                               |
|----------|--------|-----------------------------------------------|
| pathName | String | The product pathName for the requested build. |
|----------|--------|-----------------------------------------------|

|                    |        |                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nike-api-caller-id | String | A platform-unique key («domain name»:«appid») that identifies the API caller to customization services. See <a href="https://cuddly-sniffle-c799ad24.pages.github.io/api-standards/#identifying-a-calling-client">Architecture Standards (https://cuddly-sniffle-c799ad24.pages.github.io/api-standards/#identifying-a-calling-client)</a> for more. |
|--------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Getting a `nike-api-caller-id`

Please contact a team member in [#nikeid-dev-systems \(https://nikedigital.slack.com/messages/C0L8C4UM7\)](https://nikedigital.slack.com/messages/C0L8C4UM7) or email [Lst-digitaltech.customization.id.systems@nike.com](mailto:Lst-digitaltech.customization.id.systems@nike.com) to obtain a key for your platform in advance of making API calls. The current list of supported values is shown below (Note: items marked with \* are for internal tracking purposes):

Table 2: Nike Platforms with Corresponding Caller IDs

| Platform                                    | Value                                  |
|---------------------------------------------|----------------------------------------|
| Nike.com Desktop PDP                        | com.nike:commerce.idpdp.desktop        |
| Nike.com Mobile PDP                         | com.nike:commerce.idpdp.mobile         |
| Nike.com Jersey PDP                         | com.nike:commerce.jerseyidpdp.desktop  |
| Nike.com Jersey Mobile PDP                  | com.nike:commerce.jerseyidpdp.mobile   |
| Nike.com Desktop 3D Builder                 | com.nike:commerce.3d.builder.desktop   |
| Nike.com Mobile 3D Builder                  | com.nike:commerce.3d.builder.mobile    |
| Converse.com Desktop PDP                    | com.converse:commerce.idpdp.desktop    |
| Converse.com Mobile PDP                     | com.converse:commerce.idpdp.mobile     |
| Converse.com Europe Desktop PDP             | com.converse.eu:commerce.idpdp.desktop |
| Converse.com Europe Mobile PDP              | com.converse.eu:commerce.idpdp.mobile  |
| Nike App iOS                                | com.nike.commerce.omega.ios            |
| Nike App Android                            | com.nike.commerce.omega.droid          |
| NikeElite.com Desktop PDP                   | com.nike.elite:commerce.idpdp.desktop  |
| NikeElite.com Mobile PDP                    | com.nike.elite:commerce.idpdp.mobile   |
| Nike Cultivator Desktop                     | com.nike.cultivator:commerce.desktop   |
| Nike Cultivator Mobile                      | com.nike.cultivator:commerce.mobile    |
| Nike Factory Collab (FC) (includes FC Lite) | com.nike.commerce.fc                   |
| Nike DOMS                                   | com.nike.commerce.doms                 |
| Nike CSP                                    | com.nike.commerce.csp                  |
| Nike Order Capture (OCP)                    | com.nike.commerce.ocp                  |
| Nike PI                                     | com.nike.commerce.pi                   |
| Nike Email                                  | com.nike.commerce.email                |
| Tmall.com (Desktop)                         | com.tmall:commerce.idpdp.desktop       |
| Tmall.com (Mobile)                          | com.tmall:commerce.idpdp.mobile        |
| WIP Tool*                                   | com.nike:wip                           |
| Test*                                       | com.nike:test                          |
| Localhost*                                  | com.nike:commerce.b16.localhost        |

## Properties With Dependencies

Table 3: Build Properties Having Dependencies

|           |        |                                                                                                                   |
|-----------|--------|-------------------------------------------------------------------------------------------------------------------|
| productId | String | Product ID for the requested build.<br>Only required when using b16Builder and passing config: builderMode: 'wip' |
| country   | String | The current country two character abbreviation, e.g. US, GB, CN. Only required when using builderProductApi       |
| locale    | String | The current locale abbreviation, e.g. en_US, en_GB, zn_CN. Only required when using builderProductApi             |

## Optional Properties

Table 4: Optional Build Properties

|                     |         |                                                                                                                                                                                                                                                                                                                                       |
|---------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| abTestCookieName    | String  | A unique user id for Optimizely to use for it's a/b tests. If anonymousId cookie is present, it will be used over the abTestCookieName.                                                                                                                                                                                               |
| builderMode         | String  | Determines which services are used by the Builder for product data and scene7 calls. In default mode, the production app uses a service call to the V4 Product service in production. In wip mode, the Builder uses the idedit preview product service and you must also supply the productId to be passed along in the service call. |
| currentEnv          | String  | Determines the environment to be used for Builder backend services. Possible values: PROD, ecnXX (any ecn environment), prdvtools1 (any preview environment). If this config value is not set, the Builder will attempt to derive it from the client host URL, and if unsuccessful will default to PROD.                              |
| dataAsStrings       | Boolean | Pass true to have the Builder API methods return data as a string, avoiding unwanted marshaling side effects.                                                                                                                                                                                                                         |
| disablePalette      | Boolean | Pass true to disable/hide the marketing components/palette navigation.                                                                                                                                                                                                                                                                |
| eMerchId            | String  | The prodigy product ID. This can be used to fetch VAS SKU data required for jerseys.                                                                                                                                                                                                                                                  |
| enableAnimationInit | Boolean | Pass true if you want to await palette item animations until builderApi.setIsVisible(true) is called on the Builder                                                                                                                                                                                                                   |
| fetchCapacity       | Boolean | Pass true to fetch capacity information along with the product.                                                                                                                                                                                                                                                                       |



|                  |              |                                                                                                                                                                                                                                                                                                  |
|------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fitToContainer   | HTML Element | When set to a DOM element (e.g <code>document.getElementById('container')</code> ), the experience will only flex to the size of that container rather than attempting to use 100vh or calculating UI dimensions based on the window.                                                            |
| hasTouch         | Boolean      | Pass <code>true</code> if the experience supports touch input.                                                                                                                                                                                                                                   |
| isColorMatching  | Boolean      | Passing <code>true</code> hides the harness menu and control buttons. This supports color matching work flow in the WIP Builder.                                                                                                                                                                 |
| metricId         | String       | Passing a <code>prebuildId</code> or <code>metricId</code> in config will force the Builder to load the build for the product and apply all the selections in the build to the loaded product. <code>metricId</code> will always be preferred over <code>prebuildId</code> if both are supplied. |
| prebuildId       | String       | Passing a <code>prebuildId</code> or <code>metricId</code> in config will force the Builder to load the build for the product and apply all the selections in the build to the loaded product. <code>metricId</code> will always be preferred over <code>prebuildId</code> if both are supplied. |
| sizeTypeRegion   | String       | The <code>sizeTypeRegion</code> corresponding to the list of <code>sizeTypes</code> that should be returned in the build data. Valid values are US, EU, JP, CN, XP.                                                                                                                              |
| viewPortSize     | String       | Sets viewport size, and creates values for <code>imageSize</code> and <code>imageQuality</code> . Can be 'S', 'M', 'L', 'XL', 'S8', 'M8', 'L8', 'XL8'.                                                                                                                                           |
| imageQuality     | Number       | Sets the PNG bit depth. Can be either 8 or 24.                                                                                                                                                                                                                                                   |
| imageSize        | Number       | Number of pixels at which the main shoe image will be rendered.                                                                                                                                                                                                                                  |
| skipPriceInfo    | Boolean      | If set to <code>true</code> , the pricing information won't be returned but style-color code will still be returned. Use for any product that is not setup in Prodigy. This flag is added to support Converse EU products.                                                                       |
| styleCode        | String       | Mandatory when <code>skipPriceInfo</code> is set to <code>true</code> . Causes the build to be saved with the <code>styleCode</code> from the input config.                                                                                                                                      |
| colorCode        | String       | Mandatory when <code>skipPriceInfo</code> is set to <code>true</code> . Causes the build to be saved with the <code>colorCode</code> from the input config.                                                                                                                                      |
| myDesignsEnabled | Boolean      | If set to <code>'true'</code> , My Designs local storage will be enabled. Default is <code>'false'</code>                                                                                                                                                                                        |

|                  |        |                                                                                                                                                                    |
|------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| myDesignCapacity | Number | Number of most recent designs to be stored in local storage. If this maximum value is exceeded, the oldest design(s) will be deleted from storage. Default is '15' |
|------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Bridge Properties

The bridge is a set of callbacks provided to the Builder config under the `config.bridge` property. They allow you to be notified by actions within the Builder.

### onAnalyticsEvent(type, payload)

Called upon user invoked events for capturing analytics data.

The `type` property will be an event name that is exposed for analytics. This list includes:

- **ANALYTICS\_APPLY\_CUSTOMIZATION**: When the user makes any customization change
- **ANALYTICS\_COMPLETE\_BUILD**: When the user presses 'done' to complete the build
- **ANALYTICS\_MARKETING\_COMPONENT\_SELECTED**: When the user selects a marketing component
- **ANALYTICS\_SELECT\_VIEW**: When the user selects a view from the carousel
- **ANALYTICS\_ZOOM\_IN**: When the user requests a higher-res image

The `payload` property contains meta information about the event type. For example, the meta information for `ANALYTICS_SELECT_VIEW` would include the selected view number. The same applies for `ANALYTICS_MARKETING_COMPONENT_SELECTED`, which will include the id and name of the marketing component.

### onApiReady(api)

Called when the API is ready, shortly after the product has loaded. This returns the API object.

**NOTE:** The build is not always ready at this point. Please use `bridge.onProductLoad(buildData)` for initial build. Then use `api.getBuild()` after that.

### onDone(buildData)

Called with the current `buildData` whenever is “done” editing their build.

### onError(error)

Called when an error response is returned from product data fetching or when a initialization error occurs on the client. Error may be either a string or a native JavaScript error.

### onPriceUpdate(priceData)

Called with price information whenever the user makes selection changes.

### onProductLoad(buildData)

Called with the current `buildData` state of the Builder whenever a new product is loaded, or a build is applied.

## Build Data

The Builder returns a `buildData` object (also referred to as Build Data), which describes the current state of the build with sizing information, style/color, current product question/answer pairs, and more. Here is a [sample buildData object \( https://nde-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html \)](https://nde-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html).

## Build Data Fields

Table 5: Fields in Build Data

| Field               | Type   | Description                                                                                                     | Example                                                                                                                                                                                                      |
|---------------------|--------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pathName            | String | The product pathName for this build.                                                                            | “ER2teamSP19_barca”                                                                                                                                                                                          |
| productId           | String | The product ID for this build.                                                                                  | “PROD372041”                                                                                                                                                                                                 |
| consumerQuesAnswers | Array  | Collection of product question-and-answer pairs representing the current state of the Builder’s pid selections. | See <a href="https://nde-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nde-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |



| Field                  | Type   | Description                                                                                                                                                                                                                                                                            | Example                                                                                                                                                                                                |
|------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| productQuesAnswers     | Array  | A collection of product question-and-answer pairs representing the current state of the Builder’s combination selections.                                                                                                                                                              | See <a href="https://ndevportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://ndevportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| sizeMarketingComponent | Object | DEPRECATED and replaced by sizingData (see below)                                                                                                                                                                                                                                      |                                                                                                                                                                                                        |
| viewNumbers            | String | Comma-separated list of views currently being displayed by the Builder, in the order in which they are being displayed. Provides a way to template scene7 URLs for each angle of the product currently being shown within the Builder for your own product carousels or display pages. | “1,2,3,4,5,6”                                                                                                                                                                                          |
| color                  | String | The product color code for this build.                                                                                                                                                                                                                                                 | “994”                                                                                                                                                                                                  |
| price                  | String | The product price formatted as string with the currency symbol.                                                                                                                                                                                                                        | \$180                                                                                                                                                                                                  |
| rawPrice               | Number | The product price formatted as a number.                                                                                                                                                                                                                                               | 180                                                                                                                                                                                                    |
| style                  | String | The product style code.                                                                                                                                                                                                                                                                | “CK3977”                                                                                                                                                                                               |
| country                | String | The current country code for this build.                                                                                                                                                                                                                                               | “US”                                                                                                                                                                                                   |
| locale                 | String | The current locale code for this build.                                                                                                                                                                                                                                                | “en_US”                                                                                                                                                                                                |
| imageNumbers           | String | Comma-separated list of views currently being displayed by the Builder, in the order in which they are being displayed. Provides a way to template scene7 URLs for each angle of the product for your own product carousels or display pages.                                          | See <a href="https://ndevportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://ndevportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| imageService           | String | The host of the service call used to request images of the product represented by the buildData. Use in conjunction with the viewUriTemplate field to build scene7 image service request URLs.                                                                                         | See <a href="https://ndevportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://ndevportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |

| Field                  | Type   | Description                                                                                                                                                                                                                                                                                                                                                                         | Example                                                                                                                                                                                                        |
|------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| imageUrlTemplate       | String | Scene7 parameterized URL for making requests for product images that are returned as opaque JPGs. The URL contains tags that must be replaced with valid values. These tags are {VIEW_NUMBER} (one of the values from imageNumbers), {IMAGE_WIDTH}, {BACKGROUND_COLOR} (in the form f5f5f5), & {JPG_QUALITY} (0 - 100).                                                             | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| imageUrlShadowTemplate | String | This URL is the same as the imageUrlTemplate except it has added parameters that will cause a shadow to be rendered underneath the image.                                                                                                                                                                                                                                           | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| viewService            | String | The host of the render service call used to request images of the product represented by the buildData. Use in conjunction with the viewUrlTemplate field to build scene7 render service request URL.                                                                                                                                                                               | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| viewUrlTemplate        | String | Scene7 parameterized URL for making requests for product images that are uncompressed PNGs with a transparent background. The URL contains a tag {VIEW_NUMBER} (one of the values from imageNumbers) that must be replaced with a valid value. You must also add a width to the end of the string in the form &wid={imageWidth} where {imageWidth} represents the width you desire. | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| viewUrlShadowTemplate  | String | This URL is the same as the viewUrlTemplate except it has added parameters that will cause a shadow to be rendered underneath the image.                                                                                                                                                                                                                                            | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| availability           | Object | The lead time in days for the product to be delivered, when isAvailable is true. When false, only an 'out of stock' message is returned.                                                                                                                                                                                                                                            | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |

| Field      | Type  | Description                                                                                      | Example                                                                                                                                                                                                        |
|------------|-------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| myDesigns  | Array | Array of locally-saved My Designs for the given pathName                                         | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| sizeTypes  | Array | A collection of sizeTypes based on the sizeTypeRegion that was requested in the config data.     | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |
| sizingData | Array | Contains arrays of gender, width, size, and size chart data. Used for “buying” and “size” tools. | See <a href="https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html">here ( https://nike-devportal-docs.niketech.com/doc/commerce/customization/buildDataExample.html )</a> |

More about **availability**

The below table describes the contents of the `availability` object:

Table 6: Fields in Availability Object

| Field                    | Type    | Description                                                               | Example                                                |
|--------------------------|---------|---------------------------------------------------------------------------|--------------------------------------------------------|
| isAvailable              | Boolean | Whether the current style-color is available to be purchased              | true                                                   |
| leadtimeUpperBoundInDays | Number  | The lead time for the product to be delivered, in days                    | 22                                                     |
| longCapacityMessage      | String  | The long message text describing the product availability and lead-time.  | “Custom-made and delivered to you in 3 weeks or less.” |
| shortCapacityMessage     | String  | The short message text describing the product availability and lead-time. | “GREAT CHOICE”                                         |

More about **sizingData**

- Render the gender, width, and size questions and answer them using the `setAnswer` and `setsizeAnswer` methods as the consumer makes selections.
- If a value for the `sizeTypeRegion` field is not provided when initializing the Builder, then `sizeTypes` will default to undefined.
- If `fetchCapacity` is set to `false` when initializing the Builder, then all sizes will show as out of stock.
- Capacity, inventory, restrictions, `sizeType` and prior answers to sizing questions are all taken into consideration when generating this collection of objects.

Sample `sizingData`:

```
[
  {
    "displayName":"Size Chart",
    "questionId":"huarRunESS1702:LTITEM8538:LTITEM8112:LTITEM8010:LTITEM380033",
    "type":"SzChart",
    "answers":
    [
      {
        "path":"/v2/sizecharts/men_footwear_default.xml",
        "answerId":"LTITEM10027",
        "type":"SzChart",
        "isDefault":true
      }
    ]
  },
  {
    "displayName":"Gender",
    "questionId":"huarRunESS1702:LTITEM8538:LTITEM8112",
    "type":"Gender",
    "answers":
    [
      {
        "displayName":"Mens",
        "questionId":"huarRunESS1702:LTITEM8538:LTITEM8112",
        "answerId":"LTITEM8010",
        "type":"Gender",
        "isDefault":true,
        "isSelected":false,
        "isRestricted":false
      },
      {
        "displayName":"Womens",
        "questionId":"huarRunESS1702:LTITEM8538:LTITEM8112",
        "answerId":"LTITEM8011",
        "type":"Gender",
        "isDefault":false,
        "isSelected":false,
        "isRestricted":false
      }
    ]
  },
  {
    "displayName":"Width",
    "questionId":"huarRunESS1702:LTITEM8538:LTITEM210042",
    "type":"Width",
    "answers":
    [
      {
        "displayName":"Narrow Fit",
        "questionId":"huarRunESS1702:LTITEM8538:LTITEM210042",
        "answerId":"LTITEM8098",
        "type":"Width",
        "isDefault":false,
        "isSelected":false,
        "isRestricted":false
      }
    ]
  },
  {
    "displayName":"Size",
    "questionId":"huarRunESS1702:LTITEM8538:LTITEM8112:LTITEM8010:LTITEM380033",
    "importantInformation":"This product runs about a half-size smaller than standard Nike sizing.",
    "type":"Sz",
    "sizetype":"euro",
    "answers":
    [
      {
```

```

    "displayName": "39",
    "questionId": "huarRunESS1702:LTITEM8538:LTITEM8112:LTITEM8010:LTITEM380033",
    "answerId": "LTITEM8128",
    "type": "Sz",
    "sizetype": "euro",
    "isDefault": false,
    "isSelected": false,
    "isAvailable": true,
    "constrains": {
      "isInStock": true,
      "hasCapacity": true,
      "isRestricted": false
    }
  }
]
}
]

```

## Working with imageServer Images

The buildData that is returned includes URLs for generating images from the **image server**. These images are opaque JPGs, as opposed to the transparent PNGs provided by the [render server](#).

- imageNumbers
- imageService
- imageUrlTemplate
- imageUrlShadowTemplate

These can be processed to generate the urls you need by adding this function to your code:

```

/**
 * Get the imageUrls for the current design from the image server
 *
 * @param {Object} buildData - the buildData
 * @param {String} backgroundColor - the background color to use i.e. ('f5f5f5')
 * @param {Integer} imageWidth - the width value of the image
 * @param {Integer} jpgQuality - the jpeg compression quality (0 - 100), I recommend 90
 * @param {Boolean} includeShadow - whether or not to include shadow
 * @returns {Array} - an array of urls
 */
getImageUrls(buildData, backgroundColor, imageWidth, jpgQuality, includeShadow) {
  const imageUrls = [];
  const imageService = buildData.imageService;
  const imageTemplate = (includeShadow ? buildData.imageUrlShadowTemplate : buildData.imageUrlTemplate);

  if (buildData && buildData.imageNumbers && imageService && imageTemplate) {
    const viewNumbers = buildData.imageNumbers.split(',');
    const baseViewUrl = `${imageService}${imageTemplate}`;

    for (let viewIndex = 0; viewIndex < viewNumbers.length; viewIndex += 1) {
      let imageUrl = baseViewUrl.replace(/{\VIEW_NUMBER}/, viewNumbers[viewIndex])
        .replace(/{\IMAGE_WIDTH}/, imageWidth)
        .replace(/{\BACKGROUND_COLOR}/, backgroundColor)
        .replace(/{\JPG_QUALITY}/, jpgQuality);

      imageUrls.push(imageUrl);
    }
  }

  return imageUrls;
}

```

## Working with viewServer Images

The buildData that is returned will include image URLs for generating images from the **render server**. These images are transparent PNGs, as opposed to the JPGs provided by the [image server](#).

- viewNumbers
- viewService
- viewUrlTemplate

- viewUrlShadowTemplate

These can be processed to generate the URLs you need by adding this function to your code:

```
/**
 * Get the viewUrls for the current design
 *
 * @param {Object} buildData - the buildData
 * @param {Integer} imageWidth - the width value of the image
 * @param {Boolean} includeShadow - whether or not to include shadow
 * @returns {Array} - an array of urls
 */
getViewUrls(buildData, imageWidth, includeShadow) { // eslint-disable-line class-methods-use-this
  const viewUrls = [];
  const viewService = buildData.viewService;
  const viewTemplate = (includeShadow ? buildData.viewUrlShadowTemplate : buildData.viewUrlTemplate);

  if (buildData && buildData.viewNumbers && viewService && viewTemplate) {
    const viewNumbers = privateData.getBuildData().viewNumbers.split(',');
    const baseViewUrl = `${viewService}${viewTemplate}`;

    for (let viewIndex = 0; viewIndex < viewNumbers.length; viewIndex += 1) {
      const viewUrl = `${baseViewUrl.replace(/{\VIEW_NUMBER}/, viewNumbers[viewIndex])}&wid=${imageWidth}`;

      viewUrls.push(viewUrl);
    }
  }

  return viewUrls;
}
```

## Methods

### applyMyDesign

Description: Loads the design associated with the included designKey into the Builder. The onProductLoad bridge method will be called in response to a successful load.

Usage:

```
builderApi.applyMyDesign(1561569431459)
```

### clearJerseyCustomization

Description: Clears the customization (un-answers questions) and returns the new build data, for jerseys exclusively.

Usage:

```
builderApi.clearJerseyCustomization()
```

### clearMessage

Description: Clears the message in the Builder.

Usage:

```
builderApi.clearMessage()
```

### deleteMyDesign

Description: Deletes the myDesign associated with the included designKey from the localStorageCache. See also [deleteMyDesigns](#)

Usage:

```
builderApi.deleteMyDesign(1561569431459)
```

### deleteMyDesigns

Description: Deletes all of the myDesigns from the localStorageCache. See also [deleteMyDesign](#)

Usage:



```
builderApi.deleteMyDesigns()
```

## getAffectedQuestionMap

Description: Return questionsMappingInfo based on the questionId

Usage:

```
/**
 * @returns {Array}
 */
builderApi.getAffectedQuestionMap()
```

## getAnswersByCode

Description: Returns answers and their matching questionId which can then be used to display an answer. Useful for tapping into specific question or answer nodes.

Usage:

```
/**
 * @returns {Array} List of normalized questions
 */
builderApi.getAnswersByCode('jersey')
```

## getAvailability

Description: Returns promise to fetch availability information for the current configuration.

Usage:

```
/**
 • @returns {Promise->Array} a promise to return availability data
 */
builderApi.getAvailability()
```

## getAvailabilityMessages

Description: Returns promise to fetch availability messages information for the current configuration.

Usage:

```
/**
 • @returns {Promise->Object} a promise to return availability messages including leadtimeUpperBoundInDays,
 longCapacityMessage, and shortCapacityMessage
 */
builderApi.getAvailabilityMessages()
```

## getBomXML

Description: Returns the Bill Of Materials (BOM) in XML format. Filters out options with blank pidText, meaning, if the user selects only customizable options and enters no text for those options, then there will be no customization returned. Returns a Promise that either:

- Resolves to a metricId, if the builder data is valid

OR

- Rejects with an error message, if the builder data is invalid

Usage:

```
builderApi.getBomXML()
```

## getBuild

Description: Returns the current build snapshot. Filters out options with blank pidText, meaning, if the user selects only customizable options and enters no text for those options, then there will be no customization returned. Also, returns the isJerseyCustomized data.

Usage:

```
/**
 * @return {Object} The current build snapshot
 */
builderApi.getBuild()
```

## getCapacity

Description: Returns promise to fetch capacity information for current or passed-in product configuration.

Usage:

```
/**
 * @returns {Promise->Object} a promise to return capacity data
 */
builderApi.getCapacity().then()
// or
builderApi.getCapacity({ country, pathName }).then()
```

## getLeadTimeMessage

Description: Returns promise to fetch lead time information for current or passed-in product configuration.

```
/**
 * @returns {Promise->Object} a promise to return leadTimeMessage data
 */
builderApi(getLeadTimeMessage)
```

## getMarketingComponents

Description: Returns a list of normalized marketing components with nested question keys. The questions index can be used to lookup nested question values.

Usage:

```
/**
 * @returns {Object} the current prouduct marketing components data
 */
builderApi.getMarketingComponents()
```

## getMyDesigns

Description: Gets all of the myDesigns from the localStorageCache. Returns an Array of myDesigns sorted from newest to oldest.

Usage:

```
/**
 * @returns {Array}
 */
builderApi.getMyDesigns()
```

## getProductColorPalette

Description: Returns a list of unique colors for the loaded builder.

Usage:

```
/**
 * @return {Array} List of selected color answers unique by hex
 */
builderApi.getProductColorPalette()
```

## getProductFillColors

Description: Returns a list of unique colors applicable to all marketing components for the loaded Builder

Usage:

```
/**
 * @returns {Array} List of selected color answers unique by hex
 */
builderApi.getProductFillColors()
```



## getQuestionsIndex

Description: Returns a normalized array of questions keyed by their ID. Effectively, this is a flattened tree of all questions contained in the product data with nested questions as keys.

Usage:

```
/**
 * @returns {Array} List of normalized questions
 */
builderApi.getQuestionsIndex()
```

Example:

```
{
  pathName:questionId: {
    id: 'pathName:questionId',
    isHidden: false,
    questions: [
      0: 'pathName:questionId:nestedQuestionId'
    ]
  }
}
```

## getSelectedColors

Description: Returns a collection of all selected colors from the current build, unique by hex(attribute).

Usage:

```
/**
 * @returns {Array} List of selected color answers unique by hex
 */
builderApi.getSelectedColors()
```

## getSelectedColorsPatterns

Description: Returns a collection of all selected colors and patterns from the current build. Colors are unique by hex(attribute) and patterns are unique by src(attribute).

Usage:

```
/**
 * @returns {Array} List of selected color answers unique by hex & pattern type questions unique by src
 */
builderApi.getSelectedColorsPatterns
```

## getSelectedPalette

Description: Returns a list of unique colors for the selected answers of the loaded builder.

Usage:

```
/**
 * @returns {Array} List of selected color answers unique by displayName
 */
builderApi.getSelectedPalette()
```

## getSelectedPatterns

Description: Returns a collection of all selected patterns from the current build, unique by src(attribute).

Usage:

```
/**
 * @returns {Array} List of selected pattern answers unique by src
 */
builderApi.getSelectedPatterns
```

## getSelectedQuestionAnswerPairs

Description: Returns a collection of all answered questions along with their answer. For most products this will return the majority of questions, as defaults are selected in B16 APIs.

Usage:

```
/**
 * @returns {Array}
 */
builderApi.getSelectedQuestionAnswerPairs
```

getUpCharges

Description: Returns an array of applicable up-charge objects corresponding to the answered questions.

Usage:

```
/**
 * @returns {Object}
 */
builderApi.getUpCharges()
```

Example:

```
[
  {
    answer: "ANSWER_ID",
    displayName: "Display Name",
    question: "QUESTION:PATH",
    upCharge: "$0"
  }
]
```

isJerseyCustomized

Description: Returns information on the jersey customization. Filters out options with blank pidText, meaning, if the user selects only customizable options and enters no text for those options, then the function will return false.

Usage:

```
builderApi.isJerseyCustomized()
```

isPidAllowed

Description: Returns true if the profanity service allowed the string (no stop word was matched). String passed must match records in the capacity service exactly. Only single strings are supported.

Usage:

```
/**
 * @returns {Promise->Boolean} a promise to return profanity result
 */
builderApi.isPidAllowed('MYPID')
```

openHighResImageUrl

Description: Returns the high-resolution image URL in a new window.

```
builderApi.openHighResImageUrl()
```

saveBuild (Deprecated)

Description: Persists build data and returns promise to send back a metric ID for that build.

Usage:

```
/**
 * @returns {Promise->String}
 */

builderApi.saveBuild()
```

saveDesign

Description: Persists the current build after validating it can be purchased. Filters out options with blank pidText, meaning, if the user selects only customizable options and enters no text for those options, then there will be no customization returned. Returns a Promise that either:

- Resolves to a metricId, if the builder data is valid

OR

- Rejects with an error message, if the builder data is invalid

Usage:

```
/**
 * @returns {Promise->String}
 */

builderApi.saveDesign()
```

## saveMyDesign

Description: Saves a myDesign to the localStorageCache. Returns the new Array of myDesigns sorted from newest to oldest.

Usage:

```
/**
 * @returns {Array}
 */

builder.Api.saveMyDesign
```

## setAnswer

Description: Returns the current build snapshot after changing the answer to a question in the Builder. Send the full question path, e.g. "af1High14\_July:LTITEM8170:LTITEM12012:LTITEM236023:LTITEM213006" and the ID of the answer you would like to apply for that question, e.g. "LTITEM167036".

Usage:

```
/**
 * @param {String} questionId
 * @param {String} answerId
 * @param {String} pidValue
 * @returns {Object} Updated build snapshot
 */

builderApi.setAnswer(questionId, answerId, pidValue)
```

## setBuild

Description: Allows reloading the Builder with a new product by passing one of the following:

Table 7: Parameters for setBuild Method

| Param      | Type   | Result                                                                            | Example                              |
|------------|--------|-----------------------------------------------------------------------------------|--------------------------------------|
| metricId   | String | Build is loaded and applied to the product data.                                  | setBuild({ metricId: 123456789 });   |
| prebuildId | String | Build is loaded and applied to the product data.                                  | setBuild({ prebuildId: 123456789 }); |
| buildData  | Object | Product for that build is reloaded and has the build data applied to the product. | setBuild({ buildData: buildData });  |

After calling this method the bridge.onProductLoad callback is called with the buildData for the newly-applied build.

Usage:

```
/**
 * @param {Object} buildData
 */

builderApi.setBuild(buildData)
```

## setIsVisible

Description: Informs the Builder whether it is visible on-screen or not. To take effect, `enableAnimationInit: true` needs to be passed in the initialization config options.

Usage:

```
builderApi.setIsVisible(true)
```

## setMessage

Description: Causes the Builder to display a message. Takes a message object consisting of the following properties:

Table 8: Properties of Message Object in `setMessage`

| Field   | Type   | Description                                                                                                                                                                                  |
|---------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| header  | String | The text displayed at the top of the message.                                                                                                                                                |
| content | String | The text displayed under the header.                                                                                                                                                         |
| type    | String | The type of message to be displayed, which controls how the message is displayed. Default is 'COUNTDOWN'. Possible values are:<br>'COUNTDOWN',<br>'COUNTDOWN_URGENT',<br>'COUNTDOWN_EXPIRED' |

Usage:

```
/**
 * @param {Object} message
 */
builderApi.setMessage(message)
```

Examples:

```
var countdownMessage = { header: 'Available for', content: '22:14:54:06', type: 'COUNTDOWN' };
builderApi.setMessage(countdownMessage);

var countdownUrgentMessage = { header: 'Available for', content: '00:00:10:06', type: 'COUNTDOWN_URGENT' };
builderApi.setMessage(countdownUrgentMessage);

var countdownExpiredMessage = { header: 'Great choice but...', content: 'This one sold out fast. Check back soon for availability. Keep customizing, save your design and share it with friends.', type: 'COUNTDOWN_EXPIRED' };
builderApi.setMessage(countdownExpiredMessage);
```

## setSizeAnswer

Description: Given a question ID and answer ID, answer the corresponding question in the build.

Usage:

```
/**
 * @param {String} questionId
 * @param {String} answerId
 * @param {String} sizeType
 * @returns {Object} Updated build snapshot
 */
setSizeAnswer(questionId, answerId, sizeType)
```

## setSizeType

Description: Allows updating the Builder’s internal state for selected size type, which updates the size, price, and color data when the Builder creates the `buildData`.

Usage:

```
/**
 * @param {String} sizeType
 * @returns {Object} Updated build snapshot
 */
builderApi.setSizeType(sizeType)
```

## shareDesign

Description: Persists the current build and returns Promise that resolves to a metricId. Unlike [saveDesign](#), with shareDesign no validation is done on the build.

## showNotification

Description: Causes the Builder to display a notification. It takes a message object consisting of the following properties:

Table 9: Properties of Message Object in showNotification Method

| Field   | Type   | Description                                                                        |
|---------|--------|------------------------------------------------------------------------------------|
| type    | String | The type of message. Only one message of a given type will be displayed.           |
| message | String | The message text.                                                                  |
| level   | String | Optional - Determines the message level. Possible values: 'alert', undefined/null. |
| title   | String | Optional - The message title.                                                      |

Usage:

```
/**
 * @param {Object} message
 */
builderApi.showNotification(message)
```

Examples:

```
var alert = { message: 'This is an alert with the orange thing', level: 'alert', type: 'some val' };
builderApi.showNotification(alert);

var titled = { message: 'This is just a plain message, but with a title', title: 'Hey yo!', type: 'some val' };
builderApi.showNotification(titled);

var plain = { message: 'just a plain old message. not that fun, sorry.', type: 'some val' };
builderApi.showNotification(plain);
```

## Contacting the Team

|                  |                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slack            | <a href="https://nikedigital.slack.com/archives/C0L8C4UM7">#nikeid-dev-systems ( https://nikedigital.slack.com/archives/C0L8C4UM7 )</a>                         |
| Confluence Space | <a href="https://confluence.nike.com/display/NIDS/NikeiD+Systems+Home">NikeiD Systems Home ( https://confluence.nike.com/display/NIDS/NikeiD+Systems+Home )</a> |
| Team Contacts    | <a href="#">Jason Mueller, Product Manager</a>                                                                                                                  |

## Document Change Log

| Summary                                                                                                 | Date       |
|---------------------------------------------------------------------------------------------------------|------------|
| Initial publish                                                                                         | 05/17/2019 |
| Added new My Designs methods, added new shareDesign, saveDesign methods, marked saveBuild as deprecated | 07/02/2019 |
| Added new methods and buildData info for availability                                                   | 07/30/2019 |
| Added new clearCustomization method                                                                     | 04/09/2020 |

| Summary                                                                                         | Date       |
|-------------------------------------------------------------------------------------------------|------------|
| Added isJerseyCustomized and getBomXML methods, updated descriptions of getBuild and saveDesign | 04/24/2020 |

## Next Steps

- [Adding Customization To Your Experience \( https://nde-devportal-docs.niketech.com/doc/commerce/customization/use-customization.html \)](https://nde-devportal-docs.niketech.com/doc/commerce/customization/use-customization.html)
- [Using Nike APIs \( https://nde-devportal-docs.niketech.com/doc/getting-started/using-nike-apis.html \)](https://nde-devportal-docs.niketech.com/doc/getting-started/using-nike-apis.html)
- [Glossary \( https://nde-devportal-docs.niketech.com/doc/commerce/reference/glossary.html \)](https://nde-devportal-docs.niketech.com/doc/commerce/reference/glossary.html)